Examiner only

**5.** *(a)* Define the *work function* of a metal. [1]

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*(b)* The work function of caesium is $3.2 \times 10^{-19}$ J. Show that the frequency of light that will eject electrons from a caesium surface with a maximum kinetic energy of $1.5 \times 10^{-19}$ J is approximately $7 \times 10^{14}$ Hz. [2]

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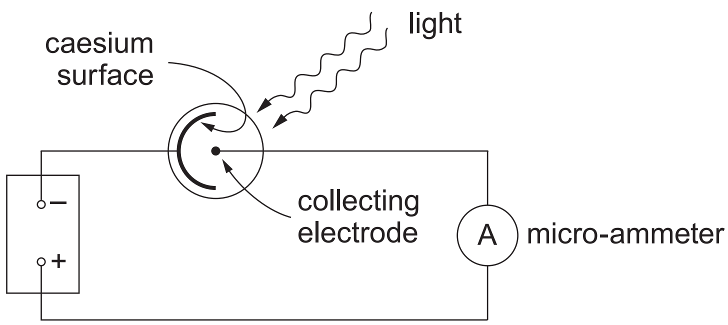
*(c)* The **same frequency** of light is shone on to the caesium surface in a vacuum photocell in the circuit shown.




The light energy falling each second on the caesium surface is 0.30 J.

(i) Show that the number of photons striking the caesium surface each second is approximately $6 \times 10^{17}$. [2]

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(ii) The current indicated by the ammeter is 0.80 µA. Calculate the number of electrons per second emitted from the caesium surface, stating your assumption. [3]

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(iii) Hence calculate the *probability* of a photon of this frequency ejecting an electron from a caesium surface. [1]

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